

Andrea Cuadra

Assistant Professor

Computer Science

Olin College of Engineering, Needham, MA

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Research Interests

Human-Computer Interaction (HCI), Human-Centered Artificial Intelligence (AI), AI Empathy, Ubiquitous Computing, Home Health, Learning Science.

Work Experience

2024–present Assistant Professor, Computer Science (HCI), Olin College

Research: Director of the Pl(ai) Lab

Teaching: HCI, Collaborative Design, Principles of Integrated Engineering

2026–2026 Visiting Professor, Key Institute in El Salvador

Research: Launched college-wide research program

Teaching: Coached faculty on project-based learning, led research seminar

2022–2024 Postdoctoral Scholar, Interaction Design Group, Stanford University

Research: Conversational interfaces

Teaching: Designing Solutions to Global Grand Challenges

Education

- 2018–2022 Ph.D., Information Science, Cornell University
Advisors: Deborah Estrin (chair), Nicola Dell (co-chair), Malte Jung (minor member), Amon Millner (external member)
Dissertation title - Inclusive Interaction Design of Voice Assistants for Older Adults
- 2018–2020 M.Sc., Information Science, Cornell University
- 2015–2017 M.Sc., Engineering: Product Design, Stanford University
Advisors: Bill Burnett and David Kelley
Design Garage Project - *Increasing Child-Parent Quality Time*: identified specific user needs, and designed solutions as part of this year-long design project. Our final deliverable was a published children’s book.
- 2009–2013 B.Sc., Engineering: Interaction Design, Olin College of Engineering
Advisor: Amon Millner
Senior Capstone Project - *Enhancing the Autodesk 123D Experience for Secondary School Students*: delivered bite-size pieces of curriculum to introduce 3D modeling software in various academic settings.
Grand Challenge Scholar Project - *Summer Camp: Motivating Students for Advanced Personalized Learning*: founded a sustainability summer camp for 8–12 year-old kids based on hands-on engineering projects and experiments.

Publications

Conference Papers (peer reviewed)

- [1] **Andrea Cuadra**, Samar Sabie, Yan Shvartzshnaider, and Deborah Estrin (2026). *Privacy Cards for Surfacing Mental Models and Exploring Privacy Concerns: A Case Study of Voice-First Ambient Interfaces with Older Adults*. Proceedings of the 2026 ACM Conference on Human Factors in Computing Systems (CHI). Article No.: 1624. (pp. 1–19). <https://doi.org/10.1145/3772318.3791893>
- [2] Alan Y. Cheng, Carolyn Q. Zou, Anthony Xie, Matthew Hsu, Felicia Yan, Felicity Huang, David Kai Zhang, Arjun Sharma, Rashon Poole, Daniel Wan Rosli, **Andrea Cuadra**, Roy Pea, and James A. Landay (2025). *Oak Story: Improving Learner Outcomes with LLM-Mediated Interactive Narratives*. Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST). (pp. 1–17). <https://doi.org/10.1145/3746059.3747698>
- [3] Andrea Green, Gabrielle Polite, Isabelle Hung, Kristen Fessele, Sarah Billington, James A. Landay, and **Andrea Cuadra** (2025). *Black Older Adults’ Perception of Using Voice Assistants to Enact a Medical Recovery Curriculum*. Proceedings of the ACM on Human-Computer Interaction. Volume 9. Issue CSCW2. Article No.: 39. (pp.1-35). <https://doi.org/10.1145/3710937>

- [4] Jianna So, Samantha Estrada, Matthew Jörke, Eva Bianchi, Maria Wang, Nava Haghighi, Kristen Fessele, James A. Landay, and **Andrea Cuadra** (2024). “*They Make Us Old Before We’re Old*”: *Designing Ethical Health Technology with and for Older Adults*. Proceedings of the ACM on Human-Computer Interaction. Volume 8. Issue CSCW2. Article No. 478: (pp 1-30). <https://doi.org/10.1145/3687017>. **Best Paper Honorable Mention Award.**
- [5] **Andrea Cuadra**, Maria Wang, Lynn Andrea Stein, Malte F. Jung, Nicola Dell, Deborah Estrin, and James A. Landay (2024). *The Illusion of Empathy? Notes on Displays of Emotion in Human-Computer Interaction*. Proceedings of the 2024 ACM Conference on Human Factors in Computing Systems (CHI). Article No.: 446. (pp 1–18). <https://doi.org/10.1145/3613904.3642336>. **Best Paper Honorable Mention Award.**
- [6] Alan Y. Cheng, Meng Guo, Melissa Ran, Arpit Ranasaria, Arjun Sharma, Anthony Xie, Khuyen N. Le, Bala Vinaithirthan, Shihe (Tracy) Luan, David Thomas Henry Wright, **Andrea Cuadra**, Roy Pea, and James A. Landay (2024). *Scientific and Fantastical: Creating Immersive, Culturally Relevant Learning Experiences with Augmented Reality and Large Language Models*. Proceedings of the 2024 ACM Conference on Human Factors in Computing Systems (CHI). Article No.: 275. (pp 1–23). <https://doi.org/10.1145/3613904.3642041>
- [7] **Andrea Cuadra**, Jessica Bethune, Alexa Lempel, Rony Krell, Katrin Hänsel, Armin Shahrokni, Deborah Estrin, and Nicola Dell (2023). *Designing Voice-First Ambient Interfaces to Support Aging in Place*. Proceedings of the 2023 Designing Interactive Systems (DIS) Conference. <https://doi.org/10.1145/3563657.3596104>
- [8] Nava Haghighi, Matthew Jörke, Yousif Mohsen, **Andrea Cuadra**, and James A. Landay (2023). *A Workshop-Based Method for Navigating Value Tensions in Collectively Speculated Worlds*. Proceedings of the 2023 DIS Conference. <https://doi.org/10.1145/3563657.3595992>
- [9] Alan Y. Cheng, Jacob Ritchie, Niki Agrawal, Elizabeth Childs, Cyan DeVeaux, Yubin Jee, Trevor Leon, Bethanie Maples, **Andrea Cuadra**, and James A. Landay (2023). *Designing Immersive, Narrative-Based Interfaces to Guide Outdoor Learning*. Proceedings of the 2023 ACM Conference on Human Factors in Computing Systems (CHI). <https://doi.org/10.1145/3544548.3581365>
- [10] **Andrea Cuadra**, Hyein Baek, Deborah Estrin, Malte F. Jung, and Nicola Dell (2022). *On Inclusion: Video Analysis of Older Adult Interactions with a Multi-Modal Voice Assistant in a Public Setting*. Proceedings of the ACM International Conference on Information & Communication Technologies and Development (ICTD). <https://doi.org/10.1145/3572334.3572371>
- [11] **Andrea Cuadra**, Shuran Li, Hansol Lee, Jason Cho, and Wendy Ju (2021). *My Bad! Repairing Intelligent Voice Assistant Errors Improves Interaction*. Proceedings of the ACM on Human-Computer Interaction. Volume 5. Issue CSCW1. Article No.: 27. (pp 1–24). <https://doi.org/10.1145/3449101>
- [12] **Andrea Cuadra**, Oluseye Bankole, and Michael Sobolev. (2021). *Planning Habit: Daily Planning Prompts with Alexa*. In *International Conference on Persuasive Technology* (pp. 73-87). Cham: Springer International. https://doi.org/10.1007/978-3-030-79460-6_7. **Shortlisted for Best Paper.**

Journals (peer reviewed)

- [1] **Andrea Cuadra**, Justine Breuch, Samantha Estrada, David Ihim, Isabelle Hung, Derek Askaryar, Marwan Hassanien, Kristen Fessele, and James A. Landay (2024). *Digital Forms for All: A Holistic Multimodal Large Language Model Agent for Health Data Entry*. ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT). Volume 8. Issue 2. Article No.: 72 (pp 1–39). <https://dl.acm.org/doi/pdf/10.1145/3659624>
- [2] **Andrea Cuadra**, Amy L. Tin, Gordon Taylor Moffat, Koshy Alexander, Robert J. Downey, Beatriz Korc-Grodzicki, Andrew J. Vickers, and Armin Shahrokni (2023). *The Association Between Perioperative Frailty and Ability to Complete a Web-Based Geriatric Assessment Among Older Adults with Cancer*. European Journal of Surgical Oncology. <https://doi.org/10.1016/j.ejso.2022.11.011>
- [3] **Andrea Cuadra**, Yen-Hao Chen, Kae-Jer Cho, Deborah Estrin, and Armin Shahrokni (2022). *Introducing the v-RFA, a voice assistant-based geriatric assessment*. Journal of Geriatric Oncology (Research Letter). <https://doi.org/10.1016/j.jgo.2022.05.001>
- [4] Jamy Li, **Andrea Cuadra**, Brian Mok, Byron Reeves, Jofish Kaye, and Wendy Ju (2019). *Communicating dominance in a nonanthropomorphic robot using locomotion*. Transactions on Human-Robot Interaction (THRI), 8(1), 4. <https://doi.org/10.1145/3310357>

Posters, Late Breaking Work, Works in Progress, and Abstracts (peer reviewed)

- [1] Yume Menghe Xu, Mark Wilson Schaffler, and **Andrea Cuadra** (2026). *Transforming Academic Machine Shops for Approachability and Equity: Identity Resources for Learning and Participation in Traditionally Masculine Spaces*. (Work in Progress). In Proceedings of the 2026 American Society for Engineering Education Annual Conference & Exposition (ASEE).
- [2] Carina Ly, Eleanor Peng, Katie Liu, Anthony Qin, Grace Howe, Alan Y. Cheng, and **Andrea Cuadra** (2025). *Museum in the Classroom: Engaging Students with Augmented Reality Museum Artifacts and Generative AI*. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA). Association for Computing Machinery, New York, NY, USA, Article 409, 1–8. <https://doi.org/10.1145/3706599.3719787>
- [3] **Andrea Cuadra**, Amy L. Tin, and Armin Shahrokni (2022). The association between perioperative frailty and ability to complete a web-based geriatric assessment in older adults with cancer. J Clin Oncol 40. (Suppl; abstr e24022). https://ascopubs.org/doi/abs/10.1200/JCO.2022.40.16_suppl.e24022
- [4] **Andrea Cuadra**, David Goedicke, and J.D. Zamfirescu-Pereira (2021). *Democratizing Design and Fabrication Using Speech: Exploring co-design with a voice assistant*. In CUI 2021 - 3rd Conference on Conversational User Interfaces (CUI), July 27–29, 2021, Bilbao (online), Spain. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3469595.3469624>
- [5] Natalie Friedman, **Andrea Cuadra**, Ruchi Patel, Shiri Azenkot, Joel Stein, and Wendy Ju (2019). *Voice Assistant Strategies and Opportunities for People with Tetraplegia*. In The 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS). ACM, New York, NY, USA, 575–577. <https://doi.org/10.1145/3308561.3354605>

Workshops Organized (peer reviewed)

- [1] Sophia Ppali, Mireia Yurrita, Alice Vitali, Alok Debnath, Lucie Flek, **Andrea Cuadra**, Sven Mayer, Michal Lahav, Tiffanie Horne, Aneesha Singh, Giulia Barbareschi, Andrea Mauri, and Himanshu Verma (2026). *(Re-)Thinking Empathy's Materiality in HCI*. Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA). Article No. 917, 8 pages. <https://doi.org/10.1145/3772363.3778717>
- [2] Elizabeth Childs, Samir Ghosh, Sebastian Cmentowski, **Andrea Cuadra**, and Rabindra Ratan (2025). *Purposeful XR: Affordances, Challenges, and Speculations for an Ethical Future*. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA). Association for Computing Machinery, New York, NY, USA, Article 795, 1–6. <https://doi.org/10.1145/3706599.3706739>

Creative Projects

- [1] **Andrea Cuadra**., Sonia Doshi, Lynn Stechschulte. (2017). Goodnight News. Self-published on Amazon.
- [2] **Andrea Cuadra**. (2016). 20 1” Metal Cubes. Personal Statements art exhibit. Stanford, CA.
- [3] **Andrea Cuadra**. (2016). Baking Blocks. Materials: Inkjet on paper (24” by 45.5”). Look Closer art exhibit by show curator Camille Utterback. Coulter Art Gallery. Stanford, CA.

Selected Honors and Awards

2025	Wake Forest Educating Character Initiative—Connecting Civic Virtue and Undergraduate Engineering (\$50,000 award, Co-PI)
2024	Best Paper Honorable Mention, ACM CSCW for [4] in conference papers (top 4%, senior author)
2024	Best Paper Honorable Mention, ACM CHI for [5] in conference papers (top 5%, first author)
2023	Stanford HAI—Personal and Private Ambient Intelligence for Senior Care Seed Grant (\$75,000 award, supporting role)
2023	NAM Healthy Longevity Catalyst Award—COBRA: Cancer, Older adults, Balance, and Resistance Activities (\$50,000 award, Co-PI)
2023	Stanford HAI—Generative AI for the Future of Learning Seed Grant (\$5,000 award, advising role)
2022	Gordon and Betty Moore Foundation 9048 (\$120,000 subaward)
2022	Stanford PRISM Baker Postdoctoral Fellow (approx. \$35,000 award)
2022	ADA Health Disparities & Diabetes Innovative Clinical or Translational Science Award (\$600,000 award, supporting role)
2020	MSKCC—Tele-Geriatric Oncology Program; Assessment, and Care at Home for the Most Vulnerable Patients. (\$200,000 award, supporting role)
2019	Cornell Tech Digital Life Initiative Doctoral Fellow (\$6,000 award)
2013	National Academy of Engineering Grand Challenge Scholar
2009	Olin College four-year, full-tuition merit scholarship

Invited Panels and Talks

2026	Higher School of Innovation in Public Administration (ES-IAP) and Ministry of Economy of El Salvador, <i>Leaders in Public Administration for Institutional Transformation</i> , Invited Panelist	San Salvador, El Salvador
2025	National Academy of Medicine, Global Innovator Summit, <i>Innovator's Connect: AI and Digital Health Technology</i>	Global
2025	Aging Life Care Association, AI & Aging Symposium, <i>Exploring The Future of Care Technology with Older Adults</i>	New England, USA
2025	Memorial Sloan Kettering, Nurses Week, <i>Empowering Nursing Innovators Leveraging Human-Centered Design for Transformative Healthcare Interventions</i>	New York, NY
2024	Olin College, AI and Society Class, <i>The Illusion of Empathy in Conversational Agents</i> .	Needham, MA
2024	ACM CHI, <i>Workshop on EmpathiCH: Scrutinizing Empathy-Centric Design Beyond the Individual</i> , Panelist	Honolulu, HI
2023	Union College, CS Seminar Series, <i>Designing Voice Assistants Inclusively</i> .	Schenectady, NY
2022	Rochester Institute of Technology (RIT), <i>Designing Voice Assistants Inclusively</i> .	Rochester, New York
2022	La Salle University of Technology (ULSA), <i>Designing Voice Assistants Inclusively</i> .	León, Nicaragua
2021	Memorial Sloan Kettering, <i>Using Artificial Intelligence to Improve Healthcare Inclusion</i>	New York, NY
2014	TEDx, <i>Don't Shut Down Your Creativity</i>	Managua, Nicaragua
2014	Universidad Autónoma del Occidente, <i>The Fabrication Behind My Inventions</i>	Cali, Colombia
2014	IEEE 33rd Central American and Panama Student Conference, <i>Entrepreneurship in Digital Fabrication</i>	Managua, Nicaragua

Hackathon Awards

2016	Yahoo internal, 1st Place
2015	NASA SpaceApps Challenge, 1st Place and People's Choice Award
2014	Hasbro-A-Thon, 1st Place
2013	Hasbro-A-Thon, 2nd Place
2013	MIT Education Design-A-Thon, 2nd Place

Teaching Experience

Courses

2025	ENGR 3220: Human-Computer Interaction	Olin College
2025	ENGR 2250: Collaborative Design	Olin College
2024	ENGR 2110: Principles of Integrated Engineering	Olin College
2023	CS 377E: Designing Solutions to Global Grand Challenges	Stanford University
2017	Pre-Collegiate Studies: Design Thinking	Stanford University

Olin Pl(ai) Lab Advisees

2024	Elías López Dalla Nora (first year), Minuja Rajasinghe (first year), Jiayuan Liu (junior)
2025	Ryan Thrasher (first year), Xavier Nishikawa (first year), Jack Wei (first year), Annabelle Austin (sophomore), Elías López Dalla Nora (sophomore), Kenneth Xiong (sophomore), Esther Aduamah (sophomore), Ashley Yang (sophomore), Sohum Kothavade (junior), Jiayuan Liu (senior), Jeremy Skoler (recent grad), Cynthia Orofo (postdoc)

Pre-Olin/external mentorship (selected mentees with co-authored papers)

2023	Andrea Green , graduate (Ph.D.), Civil and Environmental Engineering, Stanford University
2023	Jianna So , graduate (Ph.D.), Computer Science, Harvard University
2023	David Ihim , graduate (M.Sc.), Computer Science, Stanford University
2023	Carina Ly , graduate (M.Sc.), Management Science and Engineering, Stanford University
2023	Maria Wang , undergraduate (B.Sc.), Product Design, Stanford University
2023	Gabrielle Polite , undergraduate (B.Sc.), Symbolic Systems, Stanford University
2023	Samantha Estrada , undergraduate (B.Sc.), Computer Science, Stanford University
2022	Marwan Hassanien , undergraduate (B.Sc.), Computer Science, Stanford University
2022	Derek Askaryar , undergraduate (B.Sc.), Computer Science, Stanford University
2022	Justine Breuch , graduate (volunteer), Software Engineering, Google
2022	Isabelle Hung , high school (volunteer), User Experience Design, Monta Vista High School

Teaching Assistantships

2018	INFO 5345, Developing and Designing Interactive Devices	Cornell Tech
2018	INFO 5345, Developing and Designing Interactive Devices	Cornell Tech
2017	From Ideas to Action	IDEO U
2017	Leading for Creativity	IDEO U
2017	Executive Education, Design Thinking for Customer Experience	INCAE
2017	ENGR 280, From Play to Innovation	Stanford University
2017	ME 120, History and Philosophy of Design	Stanford University
2017	ME 216B, Advanced Product Design: Implementation 1	Stanford University
2017	ME 110B, Digital Design Principles and Applications	Stanford University
2017	ME 110, Design Sketching	Stanford University
2016	ME 110B, Digital Design Principles and Applications	Stanford University
2016	ME 110, Design Sketching	Stanford University
2016	ME 115A, Introduction to Human Values in Design	Stanford University
2016	ME 101, Visual Thinking	Stanford University
2012	ENGR 2599A, Computing and Craft	Olin College

Academic Service

Chair Positions

- Associate Chair, Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, cycle spanning 2025 & 2026
- Associate Chair, Proceedings of the ACM on Computer Supported Cooperative Work and Social Computing (CSCW), July cycle spanning 2023 & 2024
- Session Chair, Speculative Design and Emerging Technologies, CSCW, 2024

Peer Review

- ACM CHI, 2018, 2019, 2021, 2022, 2023, 2024, 2025
- ACM IMWUT, 2023, 2024, 2025
- ACM CSCW, 2023, 2024
- International Journal of Human-Computer Interaction, 2022
- Alt.HRI at the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2022
- ACM THRI, 2021

- IEEE Robotics and Automation Letters (RA-L), 2021
- ACM DIS, 2021, 2025
- ACM International Conference on Tangible, Embedded and Embodied Interaction (TEI), 2017

Institutional Service and Leadership

- Olin College Connecting Civic Virtue and Undergraduate Engineering Program, Co-PI, 2025
- Stanford INSPIRE-CS Research Program (undergraduate research program), research advisor, 2023
- Stanford CURIS (Computer Science undergraduate research program), research advisor, 2023
- Stanford IxD Research Group RAship program, organizer and interviewer, 2022
- Stanford Smart Primer research retreat, organizer and speaker, 2022
- Olin College FRAME mentorship program, mentor to three Latinx undergrads, 2022
- Cornell Tech Maker Lab, teaching volunteer, 2018–2021
- Cornell University Graduate Students for Gender Inclusion in Computing, 2020
- Cornell University Cornell Tech PhD Admissions Committee, 2019
- Olin College alumna interviewer for undergraduate admissions, 2019

Other Community Service

- Babson College Global Health Innovation Lab final presentations, judge, 2024
- The first workshop on Healthy Interfaces (HEALTHI), program committee member, 2021
- Costa Rica Social innovation company (Barrilete), advisor, 2019–present
- Universidad Autónoma de Occidente FabLab inauguration via Fulbright, workshop lead and speaker, 2014

Program Participation

- NSF Engineering (ENG) CAREER Proposal Workshop Mock Panel Review Session, 2024
- ACM CHI Workshop on Post-growth HCI: Co-Envisioning HCI Beyond Economic Growth, 2024
- ACM CHI Workshop on CUI@CHI: Inclusive Design of CUIs Across Modalities and Mobilities, 2023
- ACM DIS Workshop on Towards Mutual Benefit: Reflecting on Artist Residencies as a Method for Collaboration, 2023
- Preparing Future Professors, mentored by Alark Joshi at the University of San Francisco, 2022–2023
- Rochester Institute of Technology Future Faculty Career Exploration Program, 2022

- Cornell University Virtual Washington D.C. Science Policy Experience, 2022
- CMD-IT Academic Careers Workshop (ACW), 2021
- ACM CHI Workshop on Conversational Agents for Health and Wellbeing, 2020

Industry Experience

Internships

2021	Sidewalk Labs, Mesa and Delve User Experience Research	New York, NY
2019	Google, Search User Experience Research	Mountain View, CA
2018	Google, Chrome User Experience Research	San Francisco, CA
2016	Yahoo, Flurry User Experience Research	Sunnyvale, CA
2012	(Fall) Hopper, User Experience Research and Design	Cambridge , MA
2012	(Summer) Hopper, Product Management	Boston, MA

Professional Jobs

2013–2020	Entrepreneur Toy and Game Inventor, Crazy Machine
2014–2015	Remote Technical Support, Modkit
2013–2014	User Experience Researcher and Designer, Hopper